



The Moral Machine

This four year long experiment by MIT explores the moral attitudes around the world for the classic trolley problem. <https://digit.in/oct19-61>



Adoption patterns

How autonomous vehicles affect the environment depends on adoption patterns and how they are used. <https://digit.in/oct19-62>

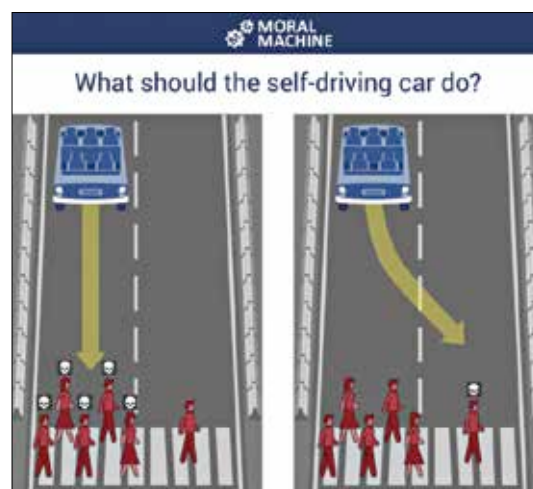


sengers, while the Chinese wanted to spare the pedestrians. What the researchers learned from the Moral Engine is that what the people want may not be the best ethical choice, and that policy makers may need to step in and make sure the vehicles take decisions that are contrary to popular expectations. For example, most people around the world felt that it was okay to prioritise the safety of a person of high status, over a person of low status.

Now the thing is, when a vehicle is programmed to act in a certain way, it is going to choose that option every time it is presented with the same choice. For example, the vehicle may be programmed to spare an 8 year old child, and hit an 80 year old man. Now, every time it is given the choice, it will do the same thing. There is no scope for the passengers to intervene in this decision. The moral choice has already been made by the manufacturers. A Chinese passenger in France may face a moral conundrum when getting into the vehicle. Passengers may be even more hesitant to get into vehicles where they know that their own lives would be given a lower priority as compared to pedestrians. One suggestion is to provide the control to the driver, if such a situation arises. However, there are studies that indicate that most people would simply not be able to react quickly enough to take such a decision, so this is not a viable approach. All of these considerations begs the question, are humans willing to let computers make life and death decisions at all?

However, considering how autonomous cars actually work, the decisions may not be so black and white, as just making a choice and going with it. The decision-making process is continuous, just as is the process of making sense from all the sensors

about the surroundings and the current state of the vehicle. Within the computer brains of autonomous cars, there are weights given to certain probabilities, and subsequent actions are taken based on these. For example, what appears to be a plain white surface can be interpreted as a wall of fog, an actual cement wall, or the back of a moving truck. Any judgement made on the nature of the feature



An experiment in the Moral Machine

Credit: MIT Technology Review

example, say the vehicle has to choose between crashing into an old person or a child. In France and Greece, the culture is such that the young would be spared. However, in China and Japan, where the elderly are more respected, the more popular choice was to spare the old person. However, the cultural attitudes are not the same for all kinds of choices. Given a choice between sparing the passengers and pedestrians, Japan and China were on opposite ends of the spectrum. The Japanese wanted to save the pas-

senger might change in moments, and in real time. This informs the decisions made by the vehicle. So actually incorporating the ethical considerations within the programming of self driving cars is a very nuanced undertaking. The vehicles may have to take decisions without complete information of their surroundings, constrained by the limits of the sensing systems on board.

ENVIRONMENTAL IMPACT

Most of the research assessing the environmental impact of autonomous vehicles has so far been inconclusive. This is because we do not know exactly what kind of changes the roll out of fully autonomous technologies will have on the behaviour and patterns of humans. Bear in mind that autonomous vehicles are different from electric vehicles, and it is not necessary that all autonomous vehicles will be electric. According to a study by University of California researchers, in a business as usual scenario where vehi-